Bio-Inspired Structural Plasticity as a Continual Learning Substrate for Curriculum Transformers

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Abstract

This work asks whether structural plasticity can be translated into a practical Transformer training recipe for curriculum-based continual learning. The model is a hybrid seq2seq Transformer with dense attention and dynamically sparse FFNs, where FFN connectivity evolves through coupled creation and pruning during a primary-to-middle-to-high school mathematical curriculum.

The main result is a proof of feasibility rather than a causal closure. The dynamic sparse substrate remains stable under sequential training with partial explicit retention replay, reaches competitive final performance, and uses approximately 3.2–3.3× fewer active FFN connections (1.64× fewer total parameters) than the dense baseline. A matched static sparse control shows that fixed FFN sparsity is already a strong baseline, especially under joint training, so the parameter-efficiency result should not be attributed to dynamic rewiring alone. The contribution of DynCL is therefore not raw sparse superiority, but stable dynamic training and an improved sequential curriculum profile relative to a fixed sparse mask.

The curriculum effect is substrate-dependent. Curriculum improves over joint training on the dense substrate, mostly redistributes performance on the dynamic substrate, and is weaker than joint training for the static sparse substrate. The tested diagonal EWC configuration provides no measurable benefit under this compositional curriculum with partial explicit retention replay. The main remaining weakness of DynCL is concentrated in `algebra__linear_1d`: corrected teacher-forcing diagnostics and a targeted rollout audit indicate that local algebraic competence is present, but free-running generation remains limited by numeric-fidelity errors.

1 Introduction

The CL problem and its domain-dependence. Catastrophic forgetting [McCloskey and Cohen, 1989] is classically framed as a universal pathology of gradient-based learning: training on a new task overwrites the weights encoding prior tasks. This framing has motivated a rich literature of countermeasures — weight regularisation [Kirkpatrick et al., 2017], progressive network expansion [Rusu et al., 2016], experience replay [Buzzega et al., 2020] — most of which were developed on benchmarks composed of *orthogonal* tasks (Split-MNIST, Permuted-MNIST, Split-CIFAR).

A subtler picture emerges in *compositional* or *hierarchical* task sequences, where later tasks build on shared computational primitives rather than replacing them with nearly orthogonal objectives.

Zhang et al. [2022] show that feature forgetting must be measured directly rather than inferred from task accuracy alone. Lee et al. [2021] show analytically that forgetting depends jointly on feature-space similarity and readout similarity: when both are aligned, sequential learning of related tasks interferes far less than the classical catastrophic-forgetting literature would predict. This suggests that later tasks may continue to reinforce shared primitives, potentially providing a form of implicit rehearsal. In the present experiments, however, that hypothesis remains confounded with partial explicit retention replay. The null result for the tested diagonal EWC configuration is therefore consistent with this interpretation, but does not prove it.

Structural plasticity as a CL mechanism. The brain circumvents the stability-plasticity dilemma not only through synaptic weight updates but through *structural* changes: synaptogenesis (creation of new connections) and synaptic pruning (elimination of unused ones) [Bhatt et al., 2009, Grutzendler et al., 2002]. Older synapses serving stable functions exhibit lower plasticity than newly formed ones, motivating an age-dependent consolidation heuristic that is only loosely analogous to parameter-importance protection in EWC, but without requiring an explicit second-order computation.

This biological principle suggests a hypothesis, not a claim of biological fidelity: a Transformer FFN whose topology dynamically evolves may (a) achieve greater parameter efficiency by allocating connections where needed, and (b) support more stable sequential learning than a fixed-topology baseline. The mechanism studied here is therefore best read as an *algorithmic translation* of growth/competition/pruning ideas under gradient-based optimisation, not as a faithful model of cortical plasticity.

The present paper should be read accordingly as a *proof-of-feasibility*: can a dynamically rewiring FFN remain stable, sparse, and competitive with dense baselines under a compositional mathematical curriculum?

Contributions.

1. **Architecture.** This paper introduces DynCL, a hybrid Transformer with dense attention, dynamically sparse FFNs, and a tri-phasic controller for coupled creation and pruning during sequential training.
2. **Evaluation.** The evaluation compares dense, dynamic, and matched static sparse FFN substrates under both sequential curriculum and joint training, with the tested diagonal EWC configuration as an explicit CL baseline.
3. **Main finding.** Dynamic sparse FFN training remains stable and competitive while using $\sim$3.2–3.3$\times$ fewer active FFN connections than the dense baseline. The matched static sparse control shows that a substantial part of this efficiency is due to sparsity itself, while DynCL improves over the fixed sparse mask in the sequential curriculum condition.
4. **Diagnosis.** The tested diagonal EWC configuration provides no measurable benefit under this compositional curriculum with partial explicit retention replay, and the main remaining weakness is concentrated in `linear_1d`, where the limiting factor is numeric rollout fidelity rather than missing local algebraic competence.

2 Background and Related Work

2.1 Dynamic Sparse Training

Dynamic sparse training (DST) alternates between forward passes on the current sparse topology and structural updates that prune low-utility connections and grow new ones. SET [Mocanu et al., 2018] applies random regrowth after magnitude-based pruning. RigL [Evci et al., 2020] replaces random regrowth with gradient-guided allocation. SNFS [Dettmers and Zettlemoyer, 2019] trains from scratch under a fixed parameter budget. This method differs from these in two respects: (i) creation and pruning are *locally coupled* within each FFN matrix, using a signal proportional to $\sqrt{s \cdot g}$ where s is a quality score and g the gap to a coverage target, providing finer control than global magnitude thresholds; and (ii) the architecture is *hybrid* — the attention sublayer is kept fully dense, with sparsity confined to the FFN, consistent with prior evidence that attention pruning is layer- and component-sensitive: although many heads can be removed after training, encoder-decoder attention is notably more sensitive to pruning in sequence transduction settings [Michel et al., 2019].

2.2 Continual Learning

Parametric regularisation. EWC [Kirkpatrick et al., 2017] penalises parameter changes weighted by the diagonal Fisher information matrix. Synaptic Intelligence (SI) [Zenke et al., 2017] uses an online path-integral approximation of a related parameter-importance signal. Both methods were validated primarily on orthogonal benchmarks (Split-MNIST, Permuted-MNIST), where tasks share no computational primitives.

Task similarity and forgetting dynamics. The relationship between task similarity and forgetting is *non-monotone* [Ramasesh et al., 2021]: intermediate similarity produces the highest interference, while orthogonal and nearly identical tasks both yield lower forgetting. Lee et al. [2021] derive this analytically, showing that the joint structure of input-feature similarity and readout similarity governs the forgetting regime. Lee et al. [2022] attribute the worst-case to a node re-use vs. node activation trade-off peaking at intermediate similarity.

Limits of Parametric Regularisation. Hiratani [2024] identify a difficult *concept-shift* regime in which input-feature similarity is high but readout similarity is low. They further show that full Fisher-metric regularisation can improve retention, whereas diagonal or Euclidean approximations are less robust to task similarity. This mathematical curriculum resembles this regime at the task-structure level: arithmetic primitives (addition, multiplication, modular division) are shared from primary through high-school, but the algebraic goals diverge substantially.

Shared representations and implicit rehearsal. Zhang et al. [2022] show that feature forgetting can remain substantial even when task-level performance looks strong, which makes it important to separate representation-level and answer-level diagnostics. In the present curriculum, one hypothesis is that some shared primitives receive sufficiently aligned gradient signal from later tasks to create a form of *implicit rehearsal*, potentially reducing the need for explicit parametric protection without eliminating forgetting altogether.

Structural and modular CL. PackNet [Mallya and Lazebnik, 2018] uses iterative pruning and freezing to carve out effectively task-specific subnetworks within a shared backbone. Progressive Neural Networks [Rusu et al., 2016] grow lateral connections from frozen prior columns. DER++ [Buzzega et al., 2020] combines replay with knowledge distillation. NISPA [Gurbuz and Dovrolis, 2022] is the closest prior work, combining sparse network training with CL on image classification, but without curriculum structure or seq2seq evaluation.

This work extends the structural direction to compositional mathematical reasoning, and provides a compute-constrained comparative case study of *structural* (dynamic topology) vs. *parametric* (EWC) protection mechanisms in a compositional curriculum setting.

2.3 Curriculum Learning

Bengio et al. [2009] formalise the intuition that presenting examples in order of increasing difficulty accelerates learning. Platanios et al. [2019] apply a competence-based curriculum to neural machine translation. Their curriculum is example-based, whereas the present one is defined externally at the level of mathematical modules. The curriculum is externally defined by the mathematical difficulty structure of the DeepMind Mathematics dataset [Saxton et al., 2019], progressing from primary (4 arithmetic modules) through middle-school (7 modules) to high-school (4 algebraic modules). This primary/middle-school/high-school decomposition is my own experimental organisation of the benchmark, not a partition defined in Saxton et al. [2019].

3 Architecture and CL Protocol

3.1 Hybrid Transformer

The model is an encoder-decoder Transformer with $d_{\text{model}} = 256$, $n_{\text{heads}} = 4$, and $L = 2$ encoder/decoder layers. The feed-forward sublayer has dimension $d_{ff} = 1024$. **Attention is kept fully dense. FFN connections are dynamic**: at any point during training, only a subset of the $d_{\text{model}} \times d_{ff}$ connections are active, with the topology evolving via the creation/pruning controller described below.

Initialisation. FFN connectivity is initialised to 30% of the full matrix, chosen uniformly at random. The attention parameters are initialised with the standard Xavier scheme and remain fully dense throughout.

3.2 Coupled Creation and Pruning

These mechanisms apply only to the dynamic FFN substrate. The dense and fixed static-sparse controls share the surrounding training and evaluation protocol, but do not use the creation/pruning controller.

Creation signal. At each forward pass, a gradient hook on the output of every FFN matmul computes a local *proxy gradient* for every weight position — active or inactive alike:

$$s_{ij} = \text{EMA}_\alpha\left(\frac{|\mathbf{x}_i| \cdot |\mathbf{g}_j|}{N_{\text{valid}}}\right), \tag{1}$$

$$d_{ij} = \text{EMA}_\alpha\left(\frac{\mathbf{x}_i \cdot \mathbf{g}_j}{N_{\text{valid}}}\right), \tag{2}$$

where $\mathbf{x}_i$ is the pre-FFN input activation at channel i, $\mathbf{g}_j = \partial\mathcal{L}/\partial\text{out}_j$ is the output-side back-propagated gradient, N_{valid} counts non-padding tokens, and α=0.20 in the canonical runs. This is not a separate dense shadow pass over inactive weights. The layer stores the pre-matmul activation $\mathbf{x}$ during the forward pass and receives the output-side gradient $\mathbf{g}$ through the same local hook during backpropagation. Because $\mathbf{x}$ and $\mathbf{g}$ are both dense before connectivity masking, s_{ij} and d_{ij} provide a proxy score for all potential FFN edges, including currently inactive ones. This follows the same broad rationale as Evci et al. [2020], but here the signal is accumulated continuously in EMA buffers rather than through periodic dense gradient passes. $s_{ij} \geq 0$ measures potential information-flow magnitude; d_{ij} (signed) measures directional coherence. The practical creation pipeline should therefore be read from *local to global*: first a candidate edge receives a local score, then that score is reweighted and filtered, then the surviving demand is aggregated at matrix level, and only then is the creation budget allocated.

In Phase 2 (Section 3.3), the **frustration score** of a candidate inactive position is:

$$f_{ij} = \max(0,\ s_{ij} - \min(s_{ij},\ |d_{ij}|))\,, \tag{3}$$

i.e. the fraction of gradient magnitude that is directionally incoherent across training examples. A high frustration score signals that multiple tasks or data distributions are competing for the same position, pulling in opposite directions — an unmet computational demand that existing connections cannot resolve.

Pruning. Synapse survival is scored as $\sigma_{ij} = \sqrt{\bar{u}_{ij} \cdot \hat{u}_{ij}}$, where $\bar{u}_{ij}$ and $\hat{u}_{ij}$ are EMA estimates of the mean and peak activation magnitude at position (i, j). Within each matrix, synapses below the 10th-percentile survival score of its mature active connections are candidates for pruning. A *late pruning cap* bounds the pruned count at 1.2× the synapses created in a recent window (floor 200, cap 600). This prevents runaway destruction of the topology in the final training phase, once new-synapse creation has plateaued and the turnover should slow down for stability rather than continue eroding the surviving structure. In Phase 2, pruning pressure on a matrix is further attenuated when that matrix still has high unmet creation demand, preventing simultaneous destruction and construction in the same region (Section 3.3).

3.3 Triphasic Training Schedule

The dynamic substrate faces a simple tension: creation signals tend to become self-reinforcing, because already active pathways produce larger activations and therefore attract further growth. This is the *need-of-the-void* bias. The controller addresses it with a triphasic schedule that separates *structural exploration* (Phase 1), *gradient-guided reinforcement* (Phase 2), and *topology consolidation* (Phase 3). The schedule is divided into three functional phases rather than tuned as a single monolithic controller: Phase 1 explores candidate FFN channels, Phase 2 refines the topology through individual growth and pruning, and Phase 3 freezes the graph so that weights can converge on the learned sparse structure. Exact phase lengths are reported in the reproducibility appendix.

Phase 1 — Structural channel probing. Phase 1 opens capacity coarsely at the channel level rather than at the single-synapse level. In an FFN, a hidden unit matters only if it has both incoming and outgoing edges, so isolated new synapses are often ineffective. Each Phase 1 event therefore creates a small *channel cluster*: up to k_{in}=3 new edges into column h of W^{up} and up to k_{out}=3 new edges from row h of W^{down} simultaneously.

Channel selection is *structural*, not gradient-based — this is the key mechanism for bypassing the need-of-the-void bias. Among hidden indices h, those below the 70th-percentile degree of currently active channels are eligible. Among eligible channels, unexplored ones are preferred via a reselection decay factor (repeated rejection of the same channel reduces its probability exponentially), ensuring diverse coverage of the hidden dimension. The per-epoch cluster budget is:

$$
\begin{aligned}
B^{(1)} &= \max\Big(2,\ \big\lfloor 16 \cdot f_{\text{cov}} \big\rceil\Big), \\
f_{\text{cov}} &= \begin{cases} 1 & \text{cov} < 0.55, \\ \max\big(0.15,\ \frac{1-\text{cov}}{0.45}\big) & \text{otherwise.} \end{cases}
\end{aligned} \tag{4}
$$

decreasing as global coverage grows. No individual pruning occurs during Phase 1.

Each new cluster enters a *probationary* state. After a minimum tenure of 25 epochs, its activity is evaluated:

$$A_h = \sqrt{\bar{u}_h^{up} \cdot \bar{u}_h^{down}}, \tag{5}$$

where $\bar{u}_h^{up}$ and $\bar{u}_h^{down}$ are mean usage scores across the cluster's up- and down-side edges respectively. Clusters are ranked against an EMA of the median activity $T_t = 0.20\, m_t + 0.80\, T_{t-1}$. If $A_h \geq T_t$: the cluster is **promoted** to the permanent topology. If $A_h < T_t$: the **entire cluster is removed atomically** (all $k_{in} + k_{out}$ edges simultaneously). This bundle-level decision ensures that only channels with verified end-to-end information flow are retained. This phase is intentionally exploratory: it opens candidate channels coarsely and keeps only those that demonstrate end-to-end activity.

Phase 2 — Individual synapse reinforcement. After the exploratory phase, creation switches to individual-synapse mode. The goal is no longer to open new channels, but to densify and refine paths that have already shown evidence of usefulness. For each candidate inactive position (i, j), the base local score is:

$$q_{ij} = s_{ij} \cdot \rho_j, \quad \rho_j = \text{clip}\left(\frac{Q_{0.75}(\text{support})}{\varepsilon + \text{support}_j},\ 0.25,\ 4.0\right), \tag{6}$$

where support_j is the accumulated usage of mature active synapses arriving at output channel j, and ρ_j is a *column-scarcity* upweight for underserved output channels. This equation is only the first stage of the effective controller. It answers the local question: *is this inactive position useful, and does it point toward an under-supported output channel?* The remaining stages progressively shift from candidate-level evidence to matrix-level allocation. In the canonical dynamic runs, a candidate must first satisfy a coherence filter ($\text{coh}_{ij} = \min(s_{ij}, |d_{ij}|)/(s_{ij}{+}\varepsilon) < 0.85$): this excludes positions already receiving strongly directionally consistent gradient signal, targeting the frustrated regions identified by Eq. 3. Surviving candidates are then filtered by a robust quality threshold based on median-plus-MAD statistics. The eligible set is pooled into a per-matrix creation-need estimate, smoothed over time, and used to modulate matrix-level budget allocation. Final position selection therefore depends on a lightweight hierarchical controller above q_{ij}, rather than on q_{ij} alone. Operationally, the controller can be read as a short cascade: *local edge score → column rebalancing → coherence and quality filters → matrix-level need → budgeted selection.* Phase 2 is therefore still driven by the same local reinforcement principle as Eq. 6, but the implemented controller adds a few stabilising gates so that creation is concentrated on coherent, underserved, and still under-capacity FFN regions. This exploitation bias is intentional: Phase 1 explores at channel scale, whereas Phase 2 consolidates and refines the pathways opened there.

Pruning is active only in this refinement phase. Pruning pressure on a matrix is *attenuated* when it still has high unmet creation demand via:

$$p_k = \frac{1}{1 + \alpha_c \cdot n_k}, \quad M_k^{\text{eff}} = M_k \cdot p_k, \tag{7}$$

where n_k is the normalised creation need of matrix k and α_c=1.0. Matrices still actively growing therefore experience reduced pruning pressure.

Phase 3 — Topology consolidation. Phase 3 freezes structural growth and stops topology updates. The graph learned during the exploratory and pruning phases is kept fixed, allowing the remaining weights to converge on a stable sparse topology.

3.4 Curriculum Continual Learning Protocol

Level transitions. When a level is completed, its checkpoint warm-starts the next level. The optimizer state is preserved (`resume_optimizer=True`) to avoid the catastrophic gradient shock associated with a cold restart (experimentally confirmed via ablation).

Dual-rate stability/plasticity protocol. A naive level transition exposes a simple tension. The learning rate that protects inherited synapses is too low for newly created ones, which appear late and have little optimizer history; the higher rate that helps new synapses adapt destabilises the inherited structure. The dual-rate mechanism resolves this by keeping one optimizer while rescaling updates for a subset of recently created synapses.

Let $g_2 \subseteq \{(\ell, i, j)\}$ denote the binary mask of synapses created during the current transfer that survived their first pruning step. Let η_t be the global learning rate at epoch t under the warm restart schedule, $\eta^\star_{\text{old}} = 1.5{\times}10^{-4}$ the target rate for inherited synapses, and $\eta^\star_{\text{new}} = 5{\times}10^{-4}$ the target rate for new synapses. The per-synapse update is then

$$
\begin{aligned}
&W \leftarrow W - \eta_t \cdot \tilde{m}_t, \qquad \tilde{g} = g \cdot \big(1 + (\gamma_t - 1)\, \mathbf{1}_{g_2}\big), \\
&\gamma_t = \max\big(1,\ \eta^\star_{\text{new}}/\eta_t\big),
\end{aligned}
$$

where g is the raw parameter gradient, $\tilde{g}$ is the rescaled gradient fed into Adam, and $\tilde{m}_t$ is the resulting Adam moment update. Synapses outside g_2 see the global η_t; synapses inside g_2 see an effective rate $\gamma_t \cdot \eta_t \approx \eta^{\star}_{\text{new}}$.

Implementation details matter for stability. First, the second-moment state v of newly inserted synapses is initialised at $\max(\text{median}(v_W), v_{\min})$ with $v_{\min} = 10^{-8}$, not at zero, to avoid the early-step instability of fresh Adam parameters under amplified gradients. Second, the mechanism is restricted to the warm-restart and plateau sub-phases of the structurally plastic part of training; at the Phase 2 to Phase 3 trigger, g_2 is merged into the global group and a single-rate regime resumes, reflecting the assumption that by Phase 3 the surviving topology should be treated as standard inventory rather than as a separate plastic pool. Third, the dense controls run without *dual-rate*: in the absence of structural creation, g_2 is empty by construction. The mechanism therefore isolates the specific asymmetry of the dynamic curriculum: inherited topology should change slowly, while fresh topology must adapt quickly.

Soft transfer mix. During training on level ℓ, the data mix follows a fixed non-adaptive schedule that always retains a non-zero share for prior levels. For example, at the primary → middle-school transition, the primary/middle-school share starts at 100/0 (pure primary), opens to 80/20 over a short warm-up buffer at the start of middle-school training, and shifts linearly to 20/80 by the final epoch. This is a form of *explicit retention replay through the training mix*, even though it does not use a separate replay buffer. A key limitation of the current protocol is therefore that it does not yet disentangle retention due to this explicit old/new mixture from retention due to compositional gradient alignment across reused primitives. The current curriculum is therefore not a strict no-replay continual learning protocol.

Scratchpad format. Multi-step modules use scratchpad targets that expose intermediate operations, while direct-answer modules use a single final answer. The intended role of the scratchpad is to keep reusable arithmetic primitives visible inside later tasks. In practice, output-length limits still force some targets into compressed `question = result` fallbacks, which remove direct supervision on those intermediate operations. The current protocol therefore still confounds three factors: explicit retention replay, scratchpad atomisation, and output-capacity limits.

4 Experimental Setup

4.1 Dataset

The experiments use a subset of the DeepMind Mathematics dataset [Saxton et al., 2019], organised into a three-level curriculum (15 modules total):

- **Primary**: 4 arithmetic modules: `add_or_sub`, `add_sub_multiple`, `mul`, and `div`.
- **Middle-school**: 7 modules combining mixed arithmetic and number theory (gcd, lcm, divisibility, place value, rounding).
- **High-school**: 4 algebraic modules (linear equations, sequences, polynomial evaluation).

The full list of modules with their format (direct-answer vs. scratchpad) and one representative example per module is given in Appendix E. Each module is evaluated primarily in the *balanced autoregressive answer-match* metric: exact-match agreement between the autoregressive final answer and the target answer, averaged uniformly across modules.

4.2 Baselines and Experimental Grid

The core comparison is organised as a 3×2 grid crossing substrate type with training protocol. The three substrates are dense FFN, dynamically sparse FFN (DynCL), and matched static sparse FFN. The two protocols are sequential curriculum training and joint training. EWC variants and no-prune ablations are added as targeted controls.

Table 1: Core experimental grid. Rows: substrate. Columns: training protocol.

	Sequential curriculum	Joint training
Dense	`denseP_42` → `denseM_42` → `denseH_42`	`dense_joint_42`
Dynamic sparse	`dynP` → `dynM` → `dynH`	`dyn_joint_42`
Static sparse	`sparseP_42` → `sparseM_42` → `sparseH_42`	`sparse_joint_42`

Table 1 summarises the core dense, dynamic, and static sparse comparison. The static sparse controls keep attention dense, use a fixed FFN mask with 629,144 active connections, and are trained under the same primary, middle, high, and joint regimes. They are designed to separate the effect of FFN sparsity from the effect of dynamic rewiring.

Methodological note. This matrix is comparative rather than strictly factorial in the classical sense. The six core cells share the same task family and high-level architecture, but they are not forced into identical optimisation conditions. In practice, the dense and dynamic substrates do not respond to the same optimiser or learning-rate recipe,[1] joint and curriculum training do not saturate at the same epoch count, and the dynamic curriculum includes a tri-phasic controller whose internal transitions are data-dependent. The experimental goal is therefore to obtain the strongest stable configuration within each regime, not to enforce iso-exposure or iso-epoch comparisons that would be artificial for these protocols.

[1] In the paper regime, the dynamic sparse substrate tolerates cosine decay well and therefore supports a long descending phase. By contrast, the fixed sparse and dense controls are much more sensitive to cosine decay: beyond a certain point, training can become unstable and collapse.

Targeted controls additionally include:

- **Dense** + EWC: `dense_EWC_H_42`, identical to the dense curriculum baseline, with EWC regularisation applied at each level transition using 8192 Fisher samples, with $\lambda = 30$ for the primary$\rightarrow$middle transition and $\lambda = 100$ for the middle$\rightarrow$high transition.
- **No-prune ablation**: `dynP_noprune_42`, identical to the dynamic curriculum baseline with pruning disabled during Phase 2, to isolate the role of the pruning mechanism.

All runs are on seed 42 unless stated otherwise. `dynP` is additionally validated on seeds 43 and 44 (Section 5.1).

4.3 Metrics

The headline metric of the paper is *balanced autoregressive answer-match accuracy* (**AR ans-em**): exact-match agreement between the model's autoregressive answer and the target answer (both canonicalised), averaged uniformly across modules. Retention is reported directly as the AR answer-match score on earlier level slices after later training. For example, the primary AR score of a high-school checkpoint measures retained primary-level performance. Final active synapse count (**Active synapses**) is reported as the parameter-efficiency metric.

The only TF quantity used in the core paper is the *answer-level* teacher-forcing exact match (**TF ans-em**), reported as a support metric for separating answer-capacity from autoregressive rollout quality. Scratchpad-format modules additionally expose AR step-level diagnostics (conceptual structure of the reasoning vs. numerical correctness of each step), but these are used only as supporting evidence. The full evaluation grid, including the exact computational definitions, canonicalisation procedure, aggregation rules, and auxiliary diagnostics, is documented in Appendix G. The corrected TF step-level diagnostics are used only as secondary evidence and only on scratchpad-format modules. For direct-answer modules, step-level TF parsing is undefined and is not used in the scientific claims. For scratchpad modules, additional diagnostic metrics are reported in Appendix G: TF conceptual-step accuracy, TF numeric-step accuracy, TF parse-failure rate, TF bypass rate, and targeted AR rollout diagnostics such as annotated conceptual-skeleton preservation and digit-level edit distance. These diagnostics are secondary and are used only to interpret failure modes, not as headline metrics.

5 Results

Table 2 gathers the quantitative backbone of the paper in a single place. The central reading is no longer that dynamic rewiring alone explains sparse competitiveness. The matched static sparse control shows that FFN sparsity itself is already a strong baseline. The most important difference between dynamic and static sparse sequential training is not the final high-school slice, but the retention profile. The dynamic substrate retains substantially more primary and middle-school performance at the end of the curriculum, whereas the static sparse curriculum loses more of the earlier levels despite using a comparable FFN budget. The main residual weakness of DynCL is concentrated in one unstable high-school algebra regime.

5.1 Primary Level: Structural Efficiency

Figure 1 shows the Pareto frontier on the primary evaluation.

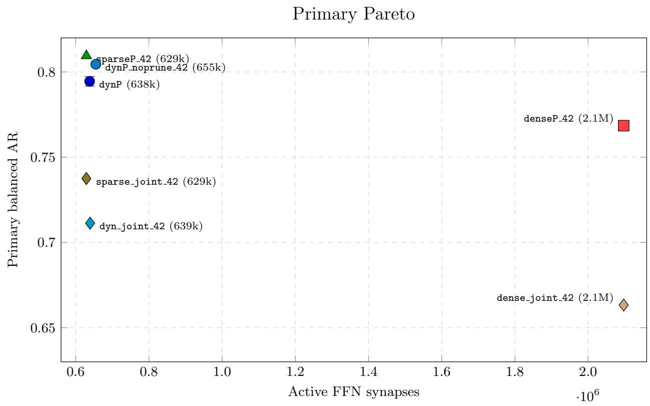

Figure 1: **Pareto frontier on the primary level.** Each point represents a run at the end of its primary training stage (or the primary slice of a joint run), plotted by active FFN synapses vs. balanced primary autoregressive answer exact match. The matched static sparse control is the strongest sparse point on this slice, while DynCL remains above the dense primary anchor with ~3.3× fewer FFN connections. Joint-training controls are shown as primary-slice references and should not be interpreted as matched primary-only training. Error bars are shown only for the multi-seed DynCL point; all other controls are single-seed.

The primary panel makes the main proof-of-feasibility claim directly visible. At the primary level, the goal is not yet to test retention, but to verify that the sparse substrates can learn the initial arithmetic slice. Both dynamic and static sparse FFNs pass this test. The best dynamic CL point (`dynP`) reaches 0.7945 with 638,087 active FFN synapses, while the dense primary anchor (`denseP_42`) reaches 0.7685 with 2,097,152 synapses. The matched static sparse control (`sparseP_42`) is even stronger at this level, reaching 0.8095 with 629,144 active FFN synapses. This does not undermine the dynamic result: it establishes that DynCL matches the fixed sparse regime at the initial level while preserving the ability to adapt its topology during later curriculum stages. The primary efficiency result must therefore be decomposed: fixed FFN sparsity is already sufficient to recover a substantial part of the performance–budget trade-off, and at least on the primary level, dynamic rewiring is not necessary to obtain a strong sparse model under this protocol. On the matched seed-42 comparison, the no-prune dy-

Table 2: Main quantitative results used throughout the paper. **AR** and **TF** denote balanced answer-level exact match. For intermediate curriculum runs, available level slices are reported; globally balanced metrics are shown only for runs that cover all three levels. Step-level diagnostics are reported only in Appendix G. Active FFN counts are end-of-run live FFN connections; total parameter counts include dense attention and embeddings.

Run	P AR	M AR	H AR	Global AR	Global TF	Active FFN	Total params	Seeds
Primary-stage runs								
`dynP`	0.7945 ± 0.0028	—	—	—	—	638k syn.	∼2.29M	42, 43, 44
`dynP_noprune_42`	0.8045	—	—	—	—	655k syn.	∼2.31M	42
`denseP_42`	0.7685	—	—	—	—	2097k syn.	∼3.75M	42
`sparseP_42`	0.8095	—	—	—	—	629k syn.	∼2.28M	42
Middle-stage curriculum runs								
`dynM`	0.7920 ± 0.0044	0.4926 ± 0.0035	—	—	—	646k syn.	∼2.30M	42, 43, 44
`denseM_42`	0.7223	0.4810	—	—	—	2097k syn.	∼3.75M	42
`dense_EWC_M_42`	0.7380	0.4783	—	—	—	2097k syn.	∼3.75M	42
`sparseM_42`	0.6755	0.4799	—	—	—	629k syn.	∼2.28M	42
Final sequential CL chains								
`dynH`‡	0.741 ± 0.004	0.489 ± 0.004	0.167 ± 0.026	0.466 ± 0.011	0.886 ± 0.002	653k syn.	∼2.31M	42, 43, 44
`denseH_42`	0.7240	0.4901	0.2517	0.489	0.881	2097k syn.	∼3.75M	42
`dense_EWC_H_42`	0.7040	0.4823	0.2507	0.479	0.881	2097k syn.	∼3.75M	42
`sparseH_42`	0.6205	0.4547	0.2490	0.4414	0.8578	629k syn.	∼2.28M	42
Joint controls								
`dyn_joint_42`	0.7113	0.4800	0.2065	0.466	0.882	639k syn.	∼2.29M	42
`dense_joint_42`	0.6633	0.4844	0.2420	0.463	0.886	2097k syn.	∼3.75M	42
`sparse_joint_42`	0.7375	0.4849	0.2355	0.4860	0.8975	629k syn.	∼2.28M	42

‡ `dynH` reports mean ± standard deviation over seeds 42, 43, 44. Static sparse rows are single-seed matched controls.

namic ablation (`dynP_noprune_42`) reaches 0.8045, compared with 0.7925 for `dynP_42`. It is also slightly above the three-seed `dynP` mean (0.7945), but that comparison mixes a single-seed ablation with a multi-seed estimate. The safest interpretation is therefore that pruning acts primarily as a structural density regulator rather than as a direct accuracy booster on the primary task.

The same panel also shows the curriculum effect inside each substrate. The dynamic joint control (`dyn_joint_42`: 0.7113) and the dense joint control (`dense_joint_42`: 0.6633) both fall substantially below their curriculum counterparts on the primary metric. At the primary level, both the curriculum and the dynamic substrate therefore provide visible gains relative to their joint controls, but the strongest sparse point is the fixed-mask control rather than the dynamic substrate. More importantly for the paper's scope, this panel shows that a dynamically rewiring FFN can remain optimisation-stable, competitive with dense baselines at much lower active FFN count, and comparable to a matched fixed sparse baseline at the initial curriculum level.

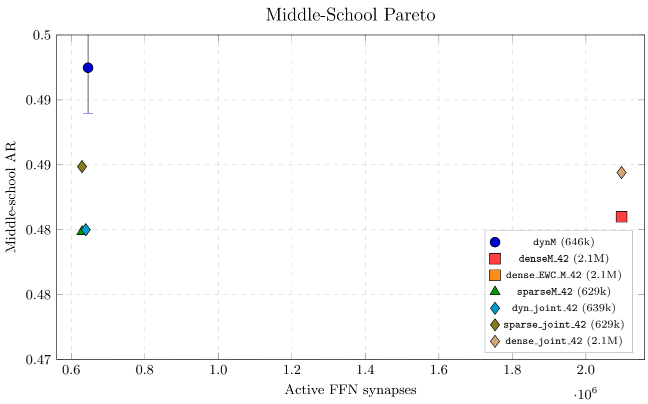

Figure 2: **Middle-school-level performance across substrates and protocols.** At the middle-school level, dense, dynamic, and static sparse models fall in a narrow AR band despite large differences in active FFN budget. Joint controls are shown as level-slice references under a different exposure regime from sequential CL. Error bars are shown only for the multi-seed DynCL point; all other controls are single-seed.

5.2 Middle-school Level: Retention Under Sequential Learning

Figure 2 shows the corresponding middle-school slice across dense, dynamic, static sparse, and joint controls. At the middle-school checkpoint, current-level AR alone is not the main continual-learning signal: the dynamic CL point (`dynM`: 0.4926 ± 0.0035), the dense CL run (`denseM_42`: 0.4810), and the matched static sparse control (`sparseM_42`: 0.4799) remain in a relatively narrow middle-school band despite large differences in active FFN budget.

A useful secondary observation is that dynamic topology does not induce large variance uniformly across the curriculum. Across seeds, DynCL is highly stable on the primary and middle-school slices, with standard deviations below one percentage point.

The retention signal is instead the primary slice after middle-school training. On this earlier slice, `dynM` retains 0.7920 ± 0.0044, whereas `sparseM_42` retains 0.6755. The dynamic curriculum therefore preserves about twelve points more primary performance than the matched static sparse curriculum while using a comparable FFN budget. This is the first point where dynamic rewiring is better interpreted as a retention mechanism rather than as a pure current-task accuracy lever.

The dense+EWC control does not alter that reading. Its middle-school score (`dense_EWC_M_42`: 0.4783) is nearly indistinguishable from `denseM_42`, and although its retained primary slice is slightly higher than the unregularised dense run (0.7380 vs. 0.7223), the overall middle-school checkpoint does not show a clear retention advantage of the tested diagonal EWC configuration under this protocol.

The joint controls are useful as reference points but should not be over-interpreted as strictly matched baselines, because they see all levels simultaneously while the CL chain measures the outcome of a sequential transfer protocol. Even with that caveat, the middle-school panel suggests a stable conclusion: by this level, the main benefit of the dynamic substrate is retention and parametric efficiency rather than a large absolute current-level AR advantage. On the dynamic substrate, curriculum learning should therefore be interpreted less as a uniform global-accuracy lever than as a way to shape the final retention profile across levels. It is useful when preserving earlier levels has value independently of the final mean score.

5.3 Effect of EWC in a Compositional Curriculum

A key motivation of this work is to characterise the role of parametric regularisation (EWC) in compositional versus orthogonal CL settings. The results show that **the tested diagonal** EWC **configuration provides no measurable benefit** under this compositional curriculum with partial explicit retention replay.

Because EWC is a retention mechanism, its relevant test is not only whether it improves the current level, but whether it preserves earlier slices after a level transition. At the primary-to-middle transition, `dense_EWC_M_42` slightly raises retained primary AR relative to `denseM_42` (0.7380 vs. 0.7223), but lowers the middle-school slice slightly (0.4783 vs. 0.4810), leaving the checkpoint interpretation essentially unchanged. At the middle-to-high transition, `dense_EWC_H_42` does not improve the retained primary and middle-school slices over `denseH_42` and does not improve the final globally balanced score. The current level scores are still shown because they remain part of the observed trade-off, but they are not the main evidence for or against EWC.

One interpretation is that this null result reflects the compositional structure of the curriculum: later mathematical tasks (middle-school modular arithmetic, high-school linear algebra) reuse the same primitive circuits (basic arithmetic, digit-level operations) as earlier ones. These shared circuits may continue to receive partially aligned gradient signal during later-level training, providing a form of *implicit rehearsal* that may make the tested diagonal Fisher penalty of EWC redundant or mildly harmful (by over-constraining parameters still in active use). Section 6.1 develops the theoretical account of this null result in detail, connecting it to the concept-shift regime of Hiratani [2024].

5.4 High-school Level and Limits

High-school summary. Dense CL (`denseH_42`: 0.252) remains above dynamic CL (`dynH`: 0.167 ± 0.026) on the current high-school slice. However, the gap is concentrated on a single module, `algebra__linear_1d`, where the dynamic seeds span $\{0.049, 0.254, 0.084\}$ in AR. On the other three high-school modules, dynamic and dense remain close. Joint controls reach `dyn_joint_42`: 0.2065 and `dense_joint_42`: 0.2420 on the same level metric.

The high-school level remains the hardest regime for all configurations. Balanced AR is 0.167 ± 0.026 for dynamic CL (three seeds) and 0.252 for dense CL (single seed). Almost all of that gap comes from `algebra__linear_1d`: the dynamic seeds score 0.049, 0.254, and 0.084 on this module, whereas dense reaches 0.372. On the three other high-school modules, dynamic and dense differ by only about one to two percentage points. The relevant reading is therefore not "dynamic fails on high-school" in general, but "the dense–dynamic gap at high-school is dominated by `linear_1d`, with strong cross-seed variance on the dynamic side".

`linear_1d` as a reliability bottleneck. The first observation is behavioural: `linear_1d` is the main source of high-school weakness, and its AR score is strongly seed-dependent. The seed that reaches `linear_1d` AR ≈ 0.25 shows that the relevant algebraic regime is reachable under the current substrate and training recipe. The limitation is therefore better described as a *reliability* or *basin-selection* problem than as a clear absence of capacity.

The second observation comes from teacher forcing. On `algebra__linear_1d`, all dynamic high-school seeds reach perfect TF answer exact match and nearly perfect TF conceptual-step accuracy, so the local algebraic procedure is available under gold-prefix conditioning even when autoregressive rollout fails.

The third observation comes from the autoregressive audit. The matched static sparse controls sharpen this reading further. `sparseH_42` and `sparse_joint_42` reach `linear_1d` AR scores of 0.368 and 0.366 respectively, substantially above the weakest dynamic seeds. The `linear_1d` failure is therefore not a generic limitation of sparse FFN reasoning. It is more consistent with a reliability interpretation: the dynamic substrate has the local algebraic competence, but some dynamic trajectories do not robustly maintain the exact numeric rollout needed for this module.

Corrected TF interpretation. The corrected by-segment teacher-forcing evaluator sharpens this interpretation substantially. Across all scratchpad-format evaluations, the evaluator yields zero TF parse failures and zero TF bypass rate, making the step-level diagnostics usable as secondary evidence. On `algebra__linear_1d`, TF numeric step accuracy remains lower than TF conceptual accuracy (0.70–0.74 for the dynamic seeds, $\sim$0.80 for dense), which is already more consistent with a numeric-rollout problem than with a missing algebraic procedure.

Manual inspection of autoregressive generations further suggests that many remaining failures are not conceptual algebraic failures but exact numeric transcription failures: large integers are copied or propagated with small digit-level corruptions. This is consistent with the gap between near-perfect TF conceptual-step accuracy and much lower AR answer accuracy. The failure mode therefore appears to involve digit fidelity and numeric copying under free-running rollout, not only missing algebraic competence.

A targeted rollout audit on 100 `algebra__linear_1d` examples per model confirms this interpretation quantitatively. Across all audited models, TF answer exact match is 1.0, TF conceptual-step accuracy is approximately 0.995–0.998, format error rate is 0.0, and the annotated conceptual skeleton is preserved in all audited examples. The remaining gap is dominated by numeric rollout fidelity: AR answer exact match ranges from 0.04 to 0.34, number-copy and operand-preservation accuracy range from 0.59 for the weakest dynamic seeds to 0.75 for the dense baseline, and numeric propagation errors remain high even when the conceptual trajectory is intact. Median final answer digit edit distance remains small (1–2 edits), indicating near-miss numeric corruptions rather than unconstrained generation failures. The stronger dynamic seed does not differ by having better TF conceptual competence, which is near-perfect for all seeds; it differs by having better numeric-copy and operand-preservation accuracy under autoregressive rollout. The per-model audit breakdown is reported in Appendix F, Table 9.

5.5 Full CL Trajectory

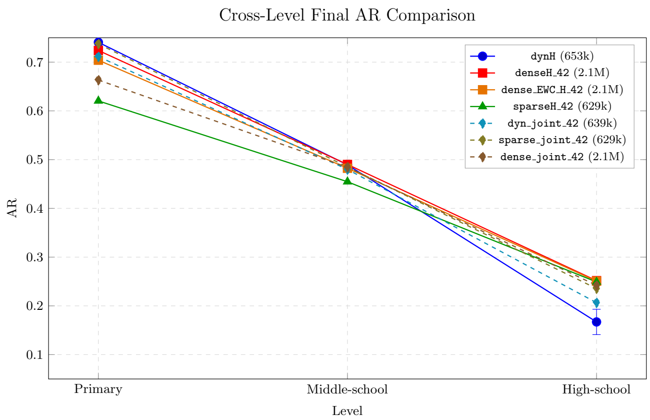

Figure 3: **Cross-level final AR comparison.** Sequential CL chains are shown as solid lines and joint controls as dashed references. For CL chains, the primary and middle-school points are end-of-high-school retention values; for joint runs, they are the corresponding level slices of the joint model. The figure highlights the substrate-dependent effect of curriculum: dense benefits globally, dynamic is approximately neutral globally but preserves a different retention profile, and static sparse favours joint training. Error bars are shown only for the multi-seed DynCL trajectory; all other controls are single-seed.

Figure 3 should be read as a final performance comparison across the three levels, rather than only as a CL-retention diagnostic. Among the sequential CL chains, the dynamic run `dynH` (seeds 42, 43, 44) tracks the dense runs closely on retained primary and middle-school while remaining much sparser than the dense substrate: end-of-training mean values are 0.741 ± 0.004 (primary), 0.489 ± 0.004 (middle-school), and 0.167 ± 0.026 (high-school). The dense and final dense+EWC cross-level profiles (single seed) are nearly identical: $0.724 \to 0.490 \to 0.252$ for `denseH_42` versus $0.704 \to 0.482 \to 0.251$ for `dense_EWC_H_42`. Here, the EWC primary and middle-school values are retained slices of the final high-school checkpoint, not separate primary-only or middle-only EWC runs. This again supports that the tested diagonal EWC configuration provides no measurable benefit under this compositional curriculum with partial explicit retention replay. At high-school, the dense–dynamic gap shows notable cross-seed variance on the dynamic side: `linear_1d` alone reaches $\{0.049, 0.254, 0.084\}$ across the three dynamic seeds, so the gap in mean values appears partly seed-driven rather than fully structural. The high-school weakness is therefore best interpreted as an unstable success regime, not as a clean failure of the dynamic substrate to represent algebraic reasoning.

The two dashed joint controls clarify a different point. `dense_joint_42` shows a smoother degradation across levels than `dyn_joint_42`, but it is not flat: performance remains highest on the primary slice and lowest on the high-school slice. `dyn_joint_42`, by contrast, trades some primary and middle-school quality for a higher final high-school slice. The static sparse controls add a third pattern: `sparseH_42` falls well below `sparse_joint_42` on the global metric (0.441 vs. 0.486), showing that a fixed sparse mask benefits much more from joint exposure than from the sequential curriculum. These joint trajectories are not retention curves, but they provide a useful reference for how simultaneous exposure redistributes AR quality across levels relative to the sequential CL chains.

Globally balanced summary. Table 3 reports a single compute-invariant summary for the runs that truly cover all three levels: AR and TF averaged uniformly across primary, middle-school and high-school. Sequential CL chains are summarised at the end of high-school training, with primary and middle-school slices being retention values; joint runs are summarised at end of joint training, with each level slice being the corresponding evaluation on that level's modules.

The balanced numbers reproduce in compact form the qualitative reading of Figure 3: `denseH_42` (0.489, single seed) leads on global AR; `dynH` (0.466 ± 0.011, three seeds) trails by $\sim$2.3 percentage points despite using $\sim$3.3$\times$ fewer active FFN connections, with a gap that lies within roughly two cross-seed standard deviations on the dynamic side and would benefit from multi-seed dense baselines to settle. The dense joint control sits slightly below both dense and dynamic CL, while the dynamic joint control is essentially tied with the dynamic CL mean

Table 3: Globally balanced AR and TF over the three curriculum levels (true 33/33/33 across primary, middle-school and high-school) for final sequential CL chains and joint controls. `dynH` is reported as mean ± standard deviation over three seeds; other rows are single-seed controls.

Run	$\mathrm{AR_{bal}}$	$\mathrm{TF_{bal}}$
Sequential CL chains		
dynH‡	0.466 ± 0.011	0.886 ± 0.002
denseH_42	0.489	0.881
dense_EWC_H_42	0.479	0.881
sparseH_42	0.441	0.858
Joint controls		
dyn_joint_42	0.466	0.882
dense_joint_42	0.463	0.886
sparse_joint_42	0.486	0.898

on the global metric. The static sparse rows make the substrate dependence sharper still: `sparse_joint_42` is a very strong global baseline at 0.486, whereas the sequential static sparse chain `sparseH_42` falls to 0.441. TF answer-level values all lie in the 0.88–0.89 band, consistent with the picture that most of the final gap is autoregressive rather than pure answer-capacity.

The dynamic sequential and dynamic joint controls therefore have nearly identical global AR, but this equality hides different cross-level profiles. Sequential DynCL retains substantially more primary and middle-school performance, whereas the dynamic joint control shifts more of its accuracy toward the high-school slice. For the dynamic substrate, curriculum is better interpreted as a retention-profile choice than as a global-accuracy lever.

5.6 Matched Static Sparse Control

All four static sparse controls are currently single-seed, keep attention dense, use a fixed sparse FFN mask with 629,144 active FFN connections, and are evaluated under the same module-balanced protocol as the dense and dynamic runs. They therefore separate *sparsity* from *dynamic rewiring* at a first level.

The resulting picture is clear. Fixed FFN sparsity is already a strong baseline. On the primary level, `sparseP_42` reaches 0.8095, above both `dynP` (0.7945) and `denseP_42` (0.7685). On the globally balanced metric, the static sparse joint control `sparse_joint_42` reaches 0.486, essentially matching the dense and dynamic global range. The parameter-efficiency result cannot therefore be attributed solely to dynamic rewiring: matched FFN sparsity alone is already sufficient to recover a substantial fraction of the sparse advantage.

At the same time, the sequential comparison remains informative. `sparseH_42` reaches 0.441 global AR, which remains below `dynH` (0.466±0.011) and `sparse_joint_42` (0.486), although the static sparse control is currently single-seed. More importantly for continual learning, the end-of-curriculum retention profile is substantially better for DynCL: at the high-school checkpoint, `dynH` retains 0.741±0.004 on the primary slice and 0.489±0.004 on the middle-school slice, whereas `sparseH_42` retains 0.6205 and 0.4547 on those slices respectively. The dynamic sparse advantage is therefore not raw sparse superiority on every slice, but a better sequential retention profile at comparable FFN scale.

5.7 Pruning as a Structural Regulator

On the matched seed-42 comparison, the no-prune ablation (`dynP_noprune_42`: 0.8045 primary AR, 655k syn.) exceeds `dynP_42` (0.7925, 638k syn.). It also sits slightly above the three-seed `dynP` mean (0.7945), but that comparison mixes a single-seed ablation with a multi-seed estimate. This indicates that pruning is not the primary driver of accuracy on the primary level; it behaves mainly as a *structural density regulator*. Without it, the FFN expands by only about 2.4% in active connections while recovering roughly one percentage point of primary AR. I therefore do not interpret pruning as an accuracy mechanism in its[2] own right, and I have no evidence in these experiments that it is the dominant driver of inter-level forgetting. The factors that did empirically govern transfer-time forgetting in our setup were preservation of the optimizer state at level hand-off (`resume_optimizer=True`) and the bounded transfer learning rate enforced by the dual-rate protocol of Section 3.4, not the pruning schedule itself.

6 Discussion

6.1 Compositional Curricula vs. EWC

The standard CL literature frames catastrophic forgetting as a universal challenge. The results suggest the framing is domain-dependent: the tested diagonal EWC configuration provides no measurable benefit — and marginally hurts — in this compositional curriculum with partial explicit retention replay, while prior work reports gains on orthogonal benchmarks [Kirkpatrick et al., 2017, Zenke et al., 2017].

The concept-shift hypothesis. Hiratani [2024] provide the most relevant theoretical account: the difficult regime is one in which input-feature similarity is high but readout goals diverge sharply — the *concept-shift* regime. Their analysis further suggests that full Fisher-metric regularisation is more robust there than diagonal or Euclidean approximations. This curriculum resembles this regime at the task-structure level: primary arithmetic primitives (addition, modular arithmetic, digit manipulation) reappear throughout middle-school and high-school problems, while the algebraic goals of the high-school level (linear equations, polynomial evaluation, sequence

[2] The current no-prune evidence is limited to the primary stage. Extending the ablation to matched middle-school and high-school chains is a priority follow-up, because those runs would directly test whether pruning contributes to inter-level retention or forgetting under the full sequential curriculum.

extrapolation) diverge substantially from the arithmetic readouts of primary. The null EWC result in this paper is therefore best interpreted more specifically as a limitation of the tested diagonal Fisher-style regularisation under this curriculum, not as a general failure of all Fisher-metric methods.

Implicit rehearsal in compositional curricula. Zhang et al. [2022] show that feature forgetting must be examined directly rather than inferred from task accuracy alone. In the present setting, one hypothesis is that some shared arithmetic primitives continue to receive aligned gradient updates from later tasks. In this setting, the basic arithmetic primitives (addition, place value, carry propagation) that are learned in primary may remain active during middle-school and high-school training, potentially constituting a form of *implicit rehearsal*. This may help explain why the tested diagonal EWC configuration does not improve the dense baseline in the current protocol: it may penalise updates to parameters that remain useful for later tasks, while potentially impeding the adaptation needed to handle new algebraic compositions. However, the current protocol still includes an explicit retention mix at later stages, so the observed retention may reflect explicit rehearsal, implicit rehearsal through reused primitives, or an interaction between both.

Implication for CL method choice. The effectiveness of parametric regularisation methods depends critically on task structure:

- *Orthogonal tasks* (Split-MNIST, Permuted-MNIST): new tasks provide little or no aligned gradient signal for old circuits; explicit protection via EWC or SI is often more appropriate and has been empirically useful in such regimes.
- *Compositional curricula*: shared primitives may receive ongoing gradient reinforcement; explicit protection may be less useful and can become overconstraining under the tested protocol.

Testing this conjecture on a truly orthogonal benchmark using the same experimental infrastructure is a direct next step. The benchmarks of Ouyang et al. [2023] for compositional CL generalization, and the NLI curriculum results of Fu et al. [2024], address similar task-structure questions, but do not directly evaluate EWC in a hierarchical mathematical curriculum.

6.2 Substrate-Dependent Curriculum Effects

The evaluation shows a substrate-dependent picture rather than a uniformly additive one. On the dense substrate, curriculum learning clearly improves over joint training across all three levels. On the dynamic substrate, the curriculum mainly redistributes quality toward retained earlier levels while leaving the final global score almost unchanged relative to the dynamic joint control. On the static sparse substrate, by contrast, joint training is stronger than the sequential curriculum at the global level (0.486 vs. 0.441). Curriculum is therefore not a uniform global-accuracy lever; its effect depends strongly on the plasticity available in the substrate.

This should not be read as a failure of the dynamic approach. Rather, the most robust result of the present study is that dynamic sparse FFN training is stable and globally competitive in a sequential compositional regime, while improving over the fixed-mask sparse curriculum condition. The precise causal role of each controller component remains to be isolated.

One plausible interpretation is that fixed sparsity is sufficient when all levels are seen jointly, because the mask can optimise for the full task mixture from the start. In the sequential curriculum, however, a fixed sparse mask must absorb new algebraic structure using the same connectivity that was shaped by earlier levels. Dynamic rewiring provides an additional adaptation channel: it can reallocate FFN capacity as the curriculum changes, which may explain why DynCL retains earlier levels better than the static sparse curriculum while remaining competitive on later levels. This also clarifies why one might have expected a stronger dynamic×curriculum interaction: if dynamic rewiring and curriculum ordering contributed largely independent retention mechanisms, their combination could in principle have produced a more than additive gain. The current results do not show such a pattern. Three non-exclusive explanations may account for the absence of multiplicative interaction:

(a) Structural protection may partially substitute for parametric protection. The dynamic substrate implements an age- and usage-based synaptic protection (older, frequently-used connections are less plastic) that plays a role loosely analogous to Fisher-weighted regularisation. If this structural mechanism partially captures the relevant importance signal, an explicit curriculum ordering provides ordinal gains (better initialisation, reduced inter-task gradient conflict) but cannot amplify the structural protection further.

(b) The compositional curriculum reduces the forgetting pressure on which the dynamic mechanism would act. If forgetting is already mild (due to shared primitives), the structural plasticity mechanism has less pathology to correct, reducing its marginal contribution in the CL regime compared to the joint regime.

(c) Architecture size as a bottleneck. At $d_{\text{model}} = 256$, $L = 2$, the model may be approaching its capacity ceiling for compositional reasoning. Synergistic effects may emerge at larger scales where each mechanism (curriculum, dynamic topology) has more room to specialise.

6.3 Limitations and Validation Roadmap

This study is a *compute-constrained proof of feasibility.* It shows that dynamically sparse FFN connectivity can be trained stably in a sequential mathematical curriculum and can approach dense performance with substantially fewer active FFN connections. What it does *not* yet provide is a full factorial validation of the underlying mechanism. The points below distinguish the current limitations from the direct next validation stage they imply.

- **Causal isolation.** The experimental grid includes a matched static sparse baseline, but this control remains single-seed and does not replace a full DST comparison. A RigL-like FFN-only sparse control at matched density, together with broader multi-seed dense and sparse validation, is still needed to separate sparsity, rewiring, and optimizer effects cleanly.
- **Replay and scratchpad confounds.** The current protocol still mixes partial explicit retention replay with scratchpad-based compositional supervision. Output-length limits can induce compressed `question = result` targets, so the present experiments cannot yet disentangle explicit replay, implicit rehearsal through reused primitives, and output-budget-induced shortcut learning.
- **Autoregressive failure analysis.** The targeted `linear_1d` audit and corrected TF diagnostics localise part of the high-school weakness to numeric rollout fidelity rather than local-competence failure. But this localisation is still partial, currently focused on one module, and should be extended with broader module-level AR analyses.
- **Scale and scope.** All experiments use a small Transformer ($d = 256$, $L = 2$) on a subset of DeepMind Mathematics. Whether the same efficiency and retention profile persists at larger scales or on more standard CL benchmarks remains open.

Next validation stage. The next validation stage is best understood as a minimal factorial design over replay and scratchpad granularity. A no-replay failure is currently ambiguous: it could mean that the implicit-rehearsal hypothesis is false, or simply that the scratchpad supervision was not sufficiently atomic because fallbacks compressed the advanced traces into input–output shortcuts. The scratchpad protocol must therefore be treated as part of the mechanism, not as a cosmetic formatting choice.

The most direct follow-up experiments are:

- **Retention-mix sweep, including a no-replay condition.** This is needed to disentangle explicit old-module replay from implicit rehearsal through compositional gradient alignment. The current old/new retention schedule ($100/0 \rightarrow 80/20 \rightarrow 20/80$ across stages) should be read as a pragmatic stability heuristic, not as a validated optimum, and it has not yet been swept. The critical comparison is a corrected atomised scratchpad with zero replay versus the same scratchpad under the current retention schedule.
- **Scratchpad granularity sweep.** The core comparison should be between (i) a corrected atomised trace with explicit binary operations and minimal fallback, (ii) a compressed-scratchpad control that still exposes intermediate reasoning but skips some primitive steps, and (iii) an answer-only extreme ablation used in appendix rather than as the main comparison. This separates compositional supervision from general sequence-to-sequence difficulty.
- **Expanded multi-seed validation.** The dynamic curriculum runs are evaluated on three seeds, but dense, static sparse, joint, EWC, and ablation controls remain single-seed unless otherwise stated. Extending these controls to multiple seeds is necessary to quantify variance robustly, especially for the high-school regime and the `algebra__linear_1d` failure mode.
- **RigL-like or SET-like sparse control.** This would test whether the structured growth, pruning, and consolidation improve over a more standard DST rewiring recipe.
- **Gradient, scratchpad, and topology diagnostics across levels.** Direct measurements of primitive reuse, gradient alignment, and topological reuse are needed to test the implicit-rehearsal hypothesis rather than inferring it only from answer-level behaviour. The key mechanistic diagnostic is gradient cosine similarity between advanced-task batches and primitive-task batches. If compositional supervision really provides implicit rehearsal, this alignment should be more positive under corrected atomised scratchpads than under compressed or answer-only traces, and more positive alignment should predict better primitive retention in the no-replay regime. It would also be valuable to analyse the learned topology itself, along with the temporal dynamics and spatial geography of connection creation and pruning as task demands evolve across curriculum levels.

Together, these experiments define the natural validation path from feasibility toward causal isolation.

7 Conclusion

This paper presented DynCL, a hybrid Transformer integrating bio-inspired ideas about structural plasticity into an algorithmic dynamic sparse FFN substrate for curriculum-based continual learning. The main result of this study is a proof of feasibility: a Transformer whose FFN topology changes during training can remain stable, sparse, and globally competitive with dense baselines under a sequential compositional mathematical curriculum.

The evaluation supports four robust empirical facts. First, the dynamic substrate remains competitive with much lower FFN parameter count ($\sim$3.2–3.3$\times$ fewer active FFN synapses). Second, the matched static sparse control shows that sparsity alone is already a strong baseline, especially under joint training. Third, the curriculum effect is substrate-dependent: it clearly improves over joint training on the dense substrate, is approximately neutral at the global level on the dynamic substrate, and is worse than joint training on the static sparse substrate. Fourth, the remaining high-school weakness is localised to `linear_1d`. Corrected TF diagnostics and the targeted AR audit show that the local algebraic procedure is available under gold-prefix conditioning; the remaining errors are dominated by numeric rollout fidelity.

On total parameter count (all components included), DynCL in its primary configuration (`dynP`, $\sim$638,000 active FFN synapses) uses $\sim$2.29M parameters versus $\sim$3.75M for the dense baseline at equivalent architecture, i.e. a $\sim$1.64$\times$ total reduction. The detailed parameter decomposition is provided in Appendix D.

A key empirical observation is that the tested diagonal EWC configuration provides no measurable benefit — and marginally hurts — under this compositional curriculum with partial explicit retention replay, where later tasks reuse shared computational primitives from earlier tasks. This is consistent with an implicit-rehearsal interpretation, while stressing that the paper does not yet provide a definitive causal decomposition between sparsity, rewiring, curriculum, and optimisation recipe.

The resulting picture is deliberately more nuanced than a simple dynamic-versus-dense comparison. Fixed FFN sparsity is already a strong baseline, especially under joint training. DynCL therefore does not show that dynamic rewiring is necessary for sparse Transformer reasoning. Its contribution is more specific: it shows that a sparse FFN substrate can be made dynamic without destabilising sequential training, and that this dynamic substrate improves the retained curriculum profile relative to a fixed sparse mask.

Thus, the present work establishes feasibility rather than mechanistic closure: DynCL shows that dynamic structural plasticity can be made stable and competitive in a Transformer-based sequential reasoning setting. The next step is a compute-intensive validation stage with expanded multi-seed controls, a matched RigL-like FFN-only DST baseline, and direct measurements of gradient alignment and topological reuse across curriculum stages. More specifically, the present results show that the dynamic substrate is stable under a curriculum with partial explicit retention replay; future work must determine whether that stability comes primarily from explicit replay, from compositional replay through corrected atomised scratchpads, or from their interaction. The critical test is therefore a corrected atomised scratchpad under zero explicit replay.

Code and artifact availability. The public reproduction repository, including code, configuration files, raw evaluation JSON files, and figure/table regeneration scripts, is available at the GitHub repository github.com/ParkerMaxime/dyncl. The paper checkpoints are archived separately on Zenodo under DOI 10.5281/zenodo.20511782. The public release is artifact-oriented: it supports figure and table reproduction from the released evaluation artifacts, but it does not expose a public checkpoint reevaluation pipeline. Full retraining is compute-intensive and is not required to reproduce the reported figures and tables from the released artifacts. Dataset usage follows the licensing and usage terms of the DeepMind Mathematics dataset.

Acknowledgements

This project was conducted independently and under constrained compute. That context shaped the methodology toward direct instrumentation, incremental ablations, and conservative interpretation of limited-seed results. I thank the open-source software and benchmarking ecosystem on which this work relies, including the release of the DeepMind Mathematics dataset.

A Experimental Context and Reproducibility Statement

Compute constraints. All experiments were conducted under a constrained compute budget. This imposed three structural constraints on the experimental design:

- *Nano-scale model.* The architecture ($d_{\text{model}} = 256$, $L = 2$, $d_{ff} = 1024$) was chosen to be trainable within the available compute envelope. Scaling experiments are deferred to future work with larger compute budgets.
- *Limited multi-seed validation.* Dynamic CL results (`dynP`, `dynM`, `dynH`) are validated on 3 seeds (42, 43, 44). Dense-baseline and joint results are single-seed in the present study; extending them to multi-seed validation is left to future work.
- *Constrained ablation coverage.* The number of ablation runs (EWC λ, pruning schedules, curriculum mix) is limited by compute availability. The ablations reported were prioritised to isolate the dominant causal factors rather than to sweep the full hyperparameter space.

Reproducibility package. To allow independent verification of all reported results, the released GitHub repository and companion Zenodo archive together provide:

- All trained checkpoints for primary, middle-school, high-school, and joint runs across dense, dynamic and static-sparse substrates, together with EWC variants and ablations for all reported seeds.
- The complete scratchpad format specification and tokenisation code used throughout training and evaluation.
- Raw evaluation JSON files containing per-module, per-metric accuracy for every reported run, allowing exact reproduction of all figures and tables from primary data.

B Run Identifiers and Naming Convention

Naming convention. Run labels follow a positional convention based on substrate, regime and curriculum level. Substrate prefixes are `dyn` (dynamic sparse FFN), `sparse` (fixed sparse FFN), and `dense` (fully dense FFN). Curriculum-level suffixes are `P` (primary), `M` (middle-school) and `H` (high-school) for sequential CL runs, and `_joint` for joint training runs that mix all three levels. Variants are appended after the level: `_EWC_` denotes addition of an Elastic Weight Consolidation regulariser, `_noprune` denotes the ablation in which Phase 2 pruning is disabled. For the dynamic joint control, `dyn_joint_42` denotes the dynamic joint-training control retained for the main comparison. Multi-seed validation status is encoded in the suffix: a bare label (e.g. `dynP`) refers to the multi-seed validated result (seeds 42, 43, 44 unless noted otherwise) and is reported as mean ± standard deviation; a numeric suffix (e.g. `dynM_42`) refers to a single specific seed.

Table 4: Run labels used throughout the paper.

Label	Level / regime	Seeds
dynP	primary, dynamic CL	42, 43, 44
dynM	middle-school, dynamic CL	42, 43, 44
dynH	high-school, dynamic CL	42, 43, 44
denseP_42	primary, dense CL	42
denseM_42	middle-school, dense CL	42
denseH_42	high-school, dense CL	42
sparseP_42	primary, static sparse CL	42
sparseM_42	middle-school, static sparse CL	42
sparseH_42	high-school, static sparse CL	42
dense_EWC_M_42	middle-school, dense CL + EWC	42
dense_EWC_H_42	high-school, dense CL + EWC	42
dense_joint_42	joint training, dense	42
dyn_joint_42	joint training, dynamic	42
sparse_joint_42	joint training, static sparse	42
dynP_noprune_42	primary, dynamic, no-pruning ablation	42

C Hyperparameters

Table 5: Main hyperparameters shared across dynamic runs.

Hyperparameter	Value
d_{model}	256
d_{ff}	1024
n_{heads}	4
$n_{\text{enc}} = n_{\text{dec}}$	2
Batch size	64
FFN init connectivity	30%
Creation budget B_{create}	400 (cap 4000)
LR schedule	warmup-plateau-cosine
LR floor ratio	0.05
Adam β_1, β_2	0.9, 0.999
Label smoothing	0.03
Max seq length (Q, A)	160, 128
Vocab size	98

Static sparse controls. Static sparse controls use the same architecture, optimiser family, curriculum ordering, batch size, training budget, dense-attention configuration, and evaluation protocol as the corresponding dynamic curriculum runs at the same level. Their learning-rate schedule follows the stable fixed-mask control recipe recorded in the released manifests, rather than the dynamic structural-plasticity schedule used by DynCL. Attention remains dense. The FFN mask is sampled once at initialisation and kept fixed throughout training, with 629,144 active FFN connections chosen to match the dynamic sparse FFN scale within approximately 1.5% of the three-seed `dynP` mean. The fixed mask uses the same per-matrix initial FFN density as the dynamic substrate (30% connectivity in each FFN matrix), but no subsequent structure updates. No dynamic creation, pruning, age, usage, tenure, or consolidation mechanism is enabled.

EWC controls. EWC variants use a diagonal Fisher estimate with 8192 Fisher samples at each level transition. The regularisation strength is $\lambda = 30$ for the primary-to-middle transition and $\lambda = 100$ for the middle-to-high transition. The Fisher estimate is computed from training samples drawn from the previously completed curriculum levels before the next-level training phase.

Training budgets and checkpoints. The headline sequential dynamic and static-sparse curriculum runs use a batch size of 64 and 750 epochs per level. Joint controls and some dense/EWC controls use run-specific budgets documented in the released configurations and manifests. Reported headline metrics are evaluated from the checkpoint specified in the released evaluation manifests. Reported active FFN counts are measured at the final checkpoint after all growth/pruning operations unless explicitly noted otherwise. All headline runs use batch size 64. The effective headline budget is therefore as follows: dynamic sequential CL and matched static sparse sequential CL both use 750 epochs per level; dense, EWC, and joint controls use run-specific learning-rate schedules, training budgets, and checkpoint kinds recorded in the released manifests. In practice, the main differences across families are the schedule recipe and the effective checkpoint retained for reporting, with dense controls more often selected from an earlier best checkpoint than from the final checkpoint.

Table 6: Effective training budgets used for headline runs.

Regime	Batch	Epochs	Checkpoint
DynCL sequential	64	750 per level	manifest checkpoint
Static sparse sequential	64	750 per level	manifest checkpoint
Dense sequential	64	run-specific	manifest checkpoint
Dense + EWC sequential	64	run-specific	manifest checkpoint
Dynamic joint	64	run-specific	manifest checkpoint
Dense joint	64	run-specific	manifest checkpoint
Static sparse joint	64	run-specific	manifest checkpoint

D Parameter Count Decomposition

This appendix decomposes the parameter count of the architecture used throughout the paper across the primary, middle-school, high-school, joint, and EWC variants. The model is a seq2seq Transformer with $L_{\text{enc}} = L_{\text{dec}} = 2$, $d_{\text{model}} = 256$, $d_{ff} = 1024$, $n_{\text{heads}} = 4$, attention dense, FFN dense or dynamic depending on the variant.

FFN block. Each FFN block contains an up-projection 256×1024 (262,144 parameters) and a down-projection 1024×256 (262,144 parameters), for 524,288 parameters per block. With 4 blocks total (2 encoder + 2 decoder), the FFN contributes $4 \times 524{,}288 = 2{,}097{,}152$ parameters when fully dense.

Attention. The architecture has 6 attention modules in total: 2 encoder self-attention, 2 decoder self-attention, and 2 decoder cross-attention. Each attention module uses 4 projections of shape 256×256, i.e. $4 \times 65{,}536 = 262{,}144$ parameters per module. Total attention parameters: $6 \times 262{,}144 = 1{,}572{,}864$.

Other components. Token embeddings, output projection (tied or untied), positional embeddings, layer norms and biases account for the residual $\sim$80,482 parameters of the dense total.

Totals by variant.

Table 7: Parameter counts for the dense, dynamic, and matched static sparse variants. "Active FFN parameters" for the dynamic variant is the count of live FFN connections at the end of primary training (`dynP`, seeds 42/43/44 mean); the static sparse variant uses a fixed FFN mask with 629,144 active connections, chosen to match the dynamic sparse FFN scale within approximately 1.5%.

Component	Dense	Dynamic	Static sparse
Attention	1,572,864	1,572,864	1,572,864
FFN	2,097,152	$\sim$638,087	629,144
Embeddings + norms	$\sim$80,482	$\sim$80,482	$\sim$80,482
Total	3,750,498	$\sim$2,291,433	$\sim$2,282,490
Total non-FFN	1,653,346	1,653,346	1,653,346

E Curriculum Modules

The 15 modules used in the curriculum are sampled from the DeepMind Mathematics dataset [Saxton et al., 2019]. They split into a *direct-answer* group (target is the answer with no intermediate reasoning) and a *scratchpad* group (target is a chain of intermediate steps separated by `|` and ending with the final answer). In the released 3C5 builders, the effective scratchpad terminator is typically `result=<value>`, although answer extraction always uses the substring after the last `=` and some historical descriptions use `answer=<value>` as a schematic placeholder. The scratchpad format used here is therefore best read as `<recopie>|<step 1>|<step 2>|...|result=<value>`. Table 8 lists the modules by level, target format, and skill probed.

Table 8: Modules per level, with format and skill probed. "D" = direct-answer target; "S" = scratchpad target with intermediate segments separated by `|`.

Level	Module	Fmt	Skill probed
P	`arithmetic__add_or_sub`	D	single + or -
	`arithmetic__add_sub_multiple`	S	chained + and -
	`arithmetic__mul`	D	integer mul
	`arithmetic__div`	D	integer div
M	`arithmetic__mixed`	S	mixed expression eval
	`numbers__div_remainder`	S	div with remainder
	`numbers__gcd`	S	Euclidean algorithm
	`numbers__lcm`	S	lcm via gcd
	`numbers__is_factor`	D	divisibility check
	`numbers__place_value`	D	digit at position
	`numbers__round_number`	D	rounding
H	`algebra__linear_1d`	S	solve a 1-var linear eq
	`algebra__sequence_next_term`	S	next term of a sequence
	`algebra__sequence_nth_term`	S	closed-form nth term
	`polynomials__evaluate`	S	evaluate a polynomial

P = Primary, M = Middle-school, H = High-school.

Target format. For scratchpad modules the target string follows the schema `recopie | step_1 | step_2 | ... | result=value`, where `recopie` is a verbatim copy of the problem statement and each subsequent step encodes a single `lhs op rhs = result` transformation. The specific operators and step types vary across modules and families (see Appendix G for the operator vocabularies used by the conceptual-signature parser).

Sampling. Headline AR/TF module-level results in the paper use held-out per-module evaluations drawn from the test split distributed with the dataset [Saxton et al., 2019], typically at 1 000 examples per module. Larger evaluation sets, up to 4 000 examples per module, are used for selected validation or diagnostic runs and are recorded in the released per-run evaluation files.

Dataset note: target fallback. Some modules, especially `arithmetic__mixed`, contain a small fraction of degenerate targets of the form `result=<value>` with no intermediate scratchpad. These cases are detected automatically and excluded from step-level aggregations to avoid biasing the step metrics. They remain included in final-answer aggregations.

Out-of-scope modules. Original DeepMind Mathematics families such as `calculus_*`, `probability_*`, and `comparison_*` are out of scope here. They are deferred to future work on broader-curriculum continual learning.

F Targeted `linear_1d` Rollout Audit

This audit uses the same fixed 100-example held-out subset for all audited models and is intended as a failure-mode diagnostic, not as a replacement for the full per-module AR score. Small differences from the full-module AR values therefore reflect the audited subsample.

For label-based audit summaries, each failed AR generation is assigned one exclusive label: `format` for malformed outputs or failed answer extraction, `early_copy` for early corruption of copied numeric state, `late_numeric` for conceptually structured traces with later numeric propagation errors, `conceptual` for wrong algebraic operations, and `other` otherwise.

Across all audited models, the annotated conceptual skeleton is preserved in every example, while median final-answer digit edit distance remains small (1–2 edits). The audit therefore supports the same interpretation as the main text: the dominant failure mode is not formatting or loss of the algebraic trajectory, but exact numeric fidelity under free-running rollout. Median first divergence appears at token position 2 for the dynamic and joint audited models, versus 5 for the dense audited model, and format error rate is 0 throughout.

G Evaluation Modes

This appendix documents the full evaluation framework used in the paper. The framework supersedes the binary "AR vs. TF" opposition that is common in earlier seq2seq literature with a finer 3×2 grid that separates *three correctness levels* from *two inference modes.*

Why a richer grid is needed

A single "teacher-forcing accuracy" label is ambiguous in the scratchpad regime. It can mean (i) token-level accuracy under the true prefix, (ii) sequence-level exact match of the full target, or (iii) final-answer match conditioned on the true scratchpad. These three quantities can differ by an order of magnitude on the same model and module. The grid below avoids that ambiguity by naming each quantity explicitly.

The grid further dissociates *procedural correctness* (the model executes the right sequence of steps) from *numerical correctness* (the model produces the right numbers in each step), which scratchpad outputs make cleanly separable.

Inference modes

AR (autoregressive). The model is given only the question and decodes the full output token-by-token, conditioning on its own previous outputs. This is the realistic inference regime.

TF (teacher-forcing, by-segment). For each scratchpad segment k, the model is given the *true* target prefix up to the start of segment k, and is then asked to generate segment k. This isolates the model's ability to produce the next step given a known-correct context. TF answers the question: "does the model have the capacity to produce this step, if the surrounding text is correct?" For direct-answer modules, TF reduces to generating the answer given the question only.

Correctness levels

For every (module, model, mode) triple, three correctness levels are reported when the module has a scratchpad. For direct-answer modules, only the answer-level metric is defined; the step-level metrics are reported as *n/a*.

1. **Final answer match (`answer_em`).** The canonicalised model answer is compared to the canonicalised target answer:

$$\mathrm{AnsMatch}_i = \mathbf{1}[\mathrm{canon}(\hat{a}_i) = \mathrm{canon}(a_i)].$$

   Aggregated as the dataset mean. This is the canonical metric of the paper. It ignores all intermediate reasoning and asks only whether the model arrives at the right final answer.

2. **Conceptual step match.** For each scratchpad segment, a *conceptual signature* $C_m(s)$ is extracted that captures the type of step performed (e.g. "copy the equation", "apply Euclidean reduction", "substitute a value", "assign the result"). The signature deliberately abstracts away from concrete numeric values. The target signature sequence and the model signature sequence are compared by an edit distance with unit insertion/deletion cost and substitution cost zero only when signatures coincide. The per-example score is

$$1 - \frac{d_C(\mathbf{c}_i, \hat{\mathbf{c}}_i)}{\max(T_i, \hat{T}_i)} \quad \text{(micro)}$$

   and the corresponding strict indicator $\mathbf{1}[d_C = 0]$ (exact). Aggregations are reported as micro-average over all steps in the dataset (main number) and as the all-or-nothing fraction (strict diagnostic). This metric answers: "does the model follow the right procedure, regardless of the numbers it produces?"

Table 9: Rollout audit on 100 `algebra__linear_1d` examples per model. TF probes local competence under gold-prefix conditioning, and AR probes free-running rollout. "Num. copy" and "Operand pres." measure exact preservation of numeric literals and operands. "Num. prop. err." denotes conceptually structured traces with incorrect numeric propagation.

Model	AR EM	TF EM	TF concept	Num. copy	Operand pres.	Num. prop. err.
`dynH_42`	0.04	1.00	0.9975	0.587	0.590	0.96
`dynH_43`	0.26	1.00	0.9950	0.677	0.680	0.74
`dynH_44`	0.05	1.00	0.9975	0.587	0.587	0.95
`denseH_42`	0.34	1.00	0.9975	0.750	0.753	0.66
`dyn_joint_42`	0.17	1.00	0.9950	0.650	0.650	0.83

3. Numeric step match. For each scratchpad segment, a *numeric signature* $N_m(s)$ is extracted that includes the canonicalised numerical values *in their roles* within the step (e.g. for a binary operation step, the canonical signature is the tuple `(binop, lhs, op, rhs, result)`). The same edit-distance machinery as above is applied, with substitution cost zero only when full numeric signatures coincide. This metric answers: "does the model produce the right numbers at each step of the procedure?"

Auxiliary diagnostics

Two additional quantities are used in the paper as supporting diagnostics:

TF parse-failure rate. Fraction of teacher-forcing evaluations for which the predicted step cannot be parsed into the expected scratchpad structure. In the corrected evaluator used in the paper, this rate is effectively zero on the analysed runs, which makes the step-level diagnostics usable as evidence.

TF bypass rate. Fraction of examples for which the target contains ≥ 2 scratchpad segments but the model collapses to a single segment instead of producing the intermediate reasoning. This is a diagnostic of scratchpad-format degradation, not a correctness metric.

Canonicalisation

A single canonicalisation function `canon(.)` is applied to both target and prediction throughout. It (i) normalises whitespace and trivial formatting variants, (ii) unifies sign and zero representations ($-0 \mapsto 0$, $+x \mapsto x$), and (iii) reduces equivalent fractions by default ($-6/8 \mapsto -3/4$).

Aggregations

Two statistics are reported wherever a step-level edit distance defines them: a *micro-average* (sum of correct steps over sum of compared steps across the dataset), and an *exact* indicator (fraction of examples for which the entire step sequence matches the target). The micro-average is the primary quantity; the exact indicator is a diagnostic of brittleness.

Across modules, level-balanced means use uniform module weights within a level. The globally balanced number quoted as "33/33/33" averages the three level means with equal weight, irrespective of the number of modules per level.

Scientific reading of the grid

The grid separates three failure profiles that the legacy "AR vs TF" opposition could not distinguish:

1. *Right procedure, wrong arithmetic.* High `concept_step_micro`, low `numeric_step_micro`, low `answer_em`. The model has learned how to reason but cannot execute the computations accurately.
2. *Wrong procedure.* Both step-level metrics low. The representation of the task itself is broken; teacher-forcing the answer no longer recovers it.
3. *Right answer via shortcut.* High `answer_em` but low step-level metrics. The model bypasses the scratchpad and produces the answer directly. Quantified by `bypass_rate`.

These three profiles correspond to different scientific questions about a model's competence and are conflated by any single-number summary.

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